

PinkyPro LCD 와 LED를 ROS2로 다루기



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Exported from Confluence · <https://pinkwink.atlassian.net>

ROS2로 LCD 다루기

[로봇] pinky_emotion 노드 실행

표정을 출력하는 서버 노드

Code

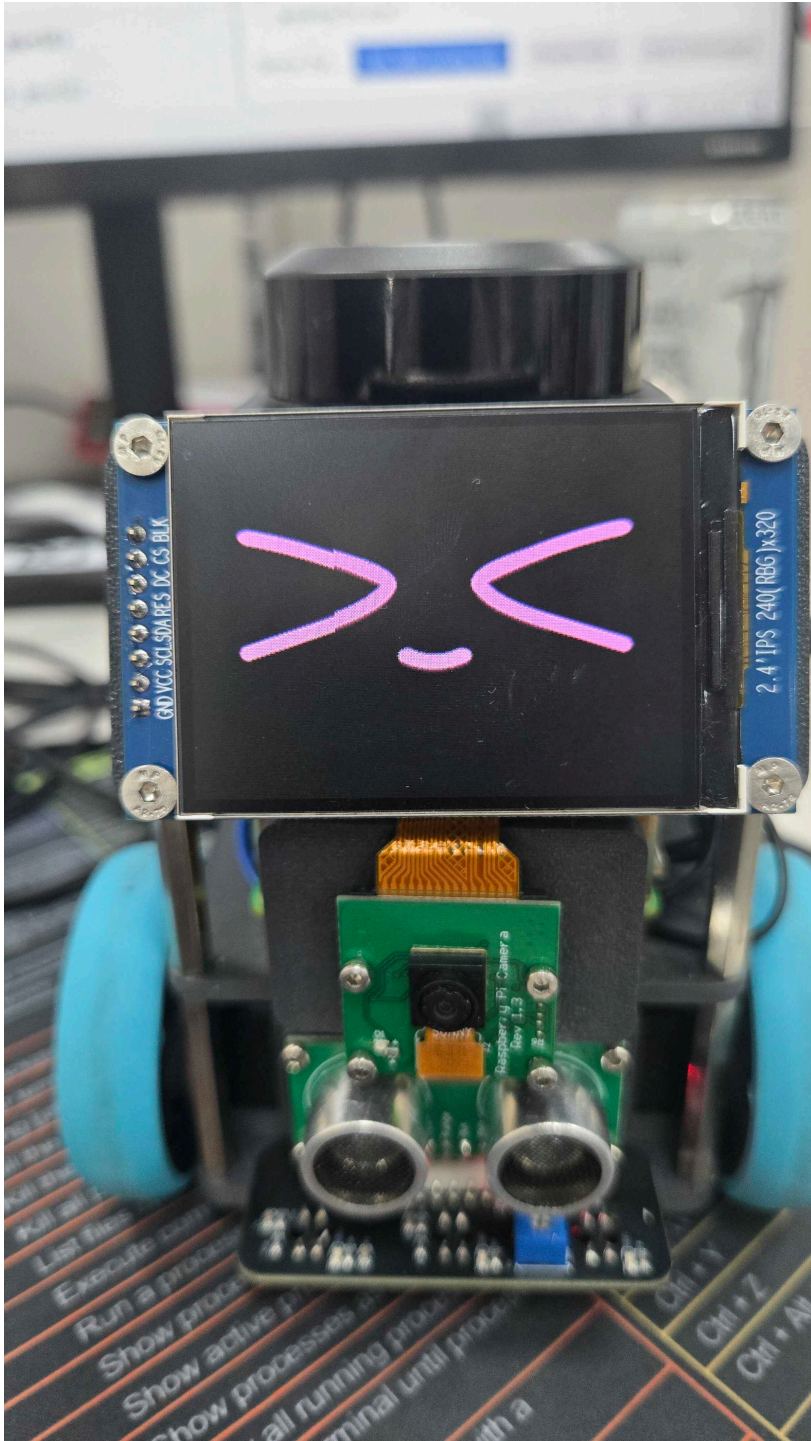
```
1 ros2 run pinky_emotion emotion_server
```

```
pinky@raspi:~$ ros2 run pinky_emotion emotion_server
[INFO] [1761033768.522564236] [pinky_emotion]: Pre-loading all emotion GIFs into memory...
[INFO] [1761033769.466823979] [pinky_emotion]: - Cached 'angry' (30 frames)
[INFO] [1761033771.933026908] [pinky_emotion]: - Cached 'basic' (80 frames)
[INFO] [1761033773.623735959] [pinky_emotion]: - Cached 'bored' (55 frames)
[INFO] [1761033774.504687219] [pinky_emotion]: - Cached 'hello' (41 frames)
[INFO] [1761033775.393514894] [pinky_emotion]: - Cached 'interest' (41 frames)
[INFO] [1761033776.124758245] [pinky_emotion]: - Cached 'fun' (38 frames)
[INFO] [1761033776.777441337] [pinky_emotion]: - Cached 'happy' (34 frames)
[INFO] [1761033778.015564733] [pinky_emotion]: - Cached 'sad' (66 frames)
[INFO] [1761033778.016324405] [pinky_emotion]: Pinky's emotion server is ready!! All GIFs pre-loaded.
```

다른 터미널에서 표정 출력 서비스로 call 명령을 보내면 LCD에 표정이 출력된다

Code

```
1 ros2 service call /set_emotion pinky_interfaces/srv/Emotion "{emotion: happy}"
```



핑키는 무려 8가지의 감정 표현이 가능하다 !

```
if emo == "hello":
    self.play_gif(self.emotion_path + "/hello.gif")

elif emo == "basic":
    self.play_gif(self.emotion_path + "/basic.gif")

elif emo == "angry":
    self.play_gif(self.emotion_path + "/angry.gif")

elif emo == "bored":
    self.play_gif(self.emotion_path + "/bored.gif")

elif emo == "fun":
    self.play_gif(self.emotion_path + "/fun.gif")

elif emo == "happy":
    self.play_gif(self.emotion_path + "/happy.gif")

elif emo == "interest":
    self.play_gif(self.emotion_path + "/interest.gif")

elif emo == "sad":
    self.play_gif(self.emotion_path + "/sad.gif")
```

ROS2로 LED 다루기

[로봇] LED 패키지 실행

Code

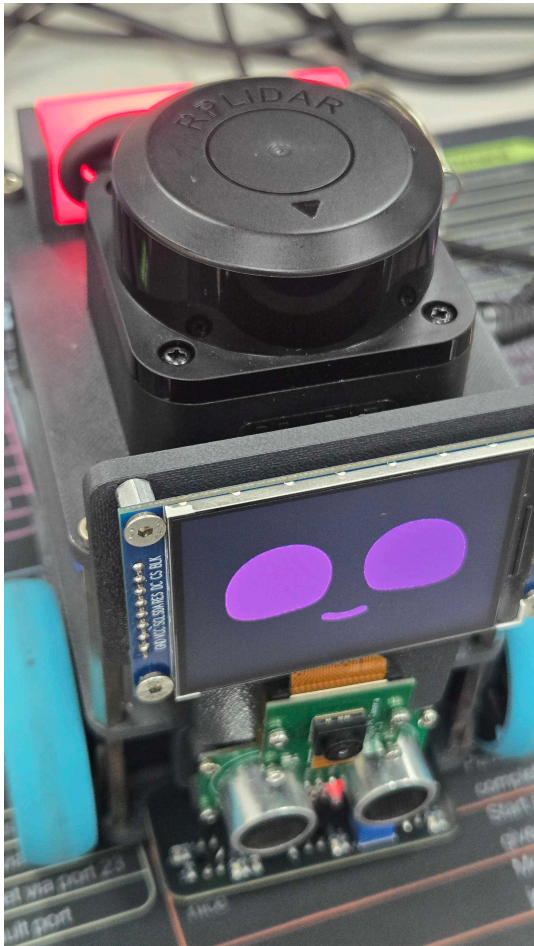
```
1  ros2 run pinky_led led_server
```

```
pinky@raspi:~$ ros2 run pinky_led led_server
[INFO] [1761033898.439458207] [led_service_server]: LED 제어 서비스 서버가 준비되었습니다.
```

LED 전체 채우기

Code

```
1  ros2 service call /set_led pinky_interfaces/srv/SetLed "{command: 'fill', r: 255, g: 0, b: 0}"
```



LED 부분 설정

Code

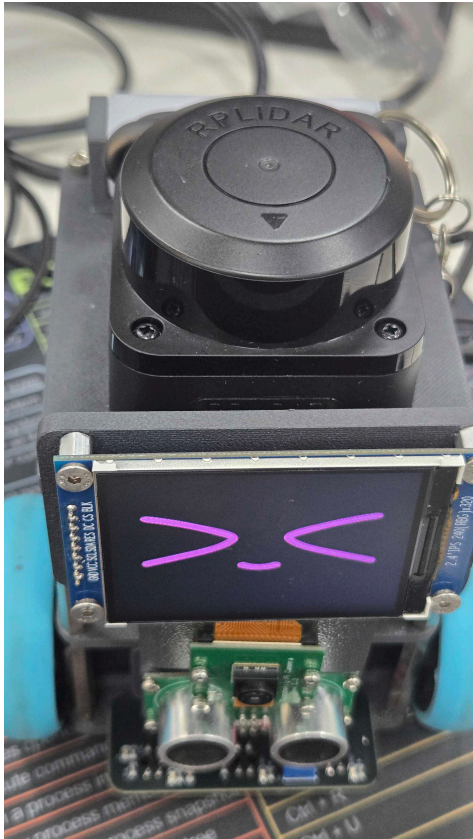
```
1  ros2 service call /set_led pinky_interfaces/srv/SetLed "{command: 'set_pixel', pixel s: [4, 5, 6, 7], r: 0, g: 0, b: 255}"
```




clear

Code

```
1  ros2 service call /set_led pinky_interfaces/srv/SetLed "{command: 'clear'}"
```



LED 밝기 설정

Code

```
1  ros2 service call /set_brightness pinky_interfaces/srv/SetBrightness "{brightness: 10}"
```

```
pinky@raspi:~$ ros2 service call /set_brightness pinky_interfaces/
srv/SetBrightness "{brightness: 10}"
waiting for service to become available...
requester: making request: pinky_interfaces.srv.SetBrightness_Requ
est(brightness=10)
```

```
response:
pinky_interfaces.srv.SetBrightness_Response(success=True, message=
'LED 밝기를 10로 설정했습니다.')
```




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